

Chapter 5

Constructing a Trade

This book has covered a number of topics but how does one tie all these concepts together and actually build a trade? Options are unique in that they have *tunable* risk-reward profiles, and the type of strategy and choice of contract parameters hugely impact the characteristics of that profile. This chapter describes some common short premium strategies and how varying each contract feature tends to alter the risk-reward properties of a short position. Some basic guidelines are also included, but the ideal trade selection ultimately depends on personal profit goals, loss tolerances, account size, and the existing positions in a portfolio. Each new trade should complement existing positions, ideally contributing some degree of diversification to the overall risk profile. However, first this chapter outlines the mechanics of building individual trades; portfolio management will be discussed later.

The general procedure for constructing a trade can be summarized as follows:

1. Choose an asset universe.
2. Choose an underlying.
3. Choose a contract duration.
4. Choose a defined or undefined risk strategy.
5. Choose a directional assumption.
6. Choose a delta.

All these factors impact the overall profile of a trade, and strategies are rarely constructed in a linear manner. Traders build trades according to their personal preferences and the size of their account, making the process of constructing a position unique. For instance, if the priority is an *undefined risk* trade, the choice of underlying will have more constraints. If the

priority is trading a *particular underlying under a certain directional assumption*, the delta and the risk definition will have more constraints.

Choose an Asset Universe

Before choosing an underlying, it's important to start with an appropriate asset universe or a set of tradable securities with desirable characteristics. The assets suitable for retail options trading must have highly liquid options markets, meaning the contracts for the security can be easily converted into cash without significantly affecting market price. To understand why liquidity is crucial, consider an example of an *illiquid* asset, such as a house. Selling a home at fair market value in a saturated housing market requires significant time and effort. Sellers run the additional risk of having to reduce the asking price significantly to secure a buyer quickly. Options illiquidity carries risk for similar reasons, and selectively trading assets with liquid options markets ensures that contract orders will be filled efficiently and at a fair market price.

Options liquidity is not equivalent to underlying liquidity. An underlying is considered liquid if it has the following characteristics:

- A high daily volume, meaning many shares traded daily (>1 million).
- A tight bid-ask spread, meaning a small difference between the maximum a buyer is willing to pay and the minimum a seller is willing to take (<0.1% of the asset price).

Some examples of liquid underlyings include AMZN, IBM, SPY, and TSLA, as shown in [Table 5.1](#) below.

Table 5.1 Pricing, bid-ask spread, and daily volume data for different equities collected on February 10, 2020, at 1 p.m.

Asset	Previous Closing Price	Bid-Ask Spread	Spread/Close (% of Closing Price)	Daily Trading Volume
AMZN	\$3,322.94	\$0.32	0.01%	1,240,935
IBM	\$121.98	\$0.05	0.04%	2,484,505
SPY	\$390.51	\$0.02	0.005%	16,619,920
TSLA	\$863.42	\$0.51	0.06%	9,371,760

It is relatively straightforward to verify underlying liquidity using daily volume and bid-ask spread as a percentage of closing price. However, a liquid underlying may not have an equally liquid options market.

Sufficiently liquid options underlyings must have *contract prices* with tight bid-ask spreads and high daily volumes. The options selection should also offer flexible time frames and strike prices. An underlying with a liquid options market is thus classified by the following properties:

- A high open interest or volume across strikes (at least a few hundred per strike).
- A tight bid-ask spread (<1% of the contract price).
- Available contracts with several strike prices and expiration dates.

Options liquidity ensures that traders have a wide selection of contracts to choose from and that short premium positions can be opened (i.e., contracts can be sold to a buyer) easily. Additionally, liquidity minimizes the risk of being stuck in a position because it allows traders to close short premium positions (i.e., identical contracts can be bought back) quickly.

The asset universe presented in this book is equity-based and mostly consists of stock and exchange-traded fund (ETF) underlyings, recalling that a stock represents a share of ownership for a single company, and an ETF tracks a specific set of securities, such as a sector, commodity, or market index. However, asset universes are generally product indifferent and can include any instruments with liquid options that present opportunities, such as commodities, digital currencies, and futures.

Choose an Underlying

The choice of underlying from a universe of sufficiently liquid assets is somewhat arbitrary, but traders often choose to trade short options on instruments for a preferred company, sector, or market under specific directional beliefs. Though this is a perfectly fine way to trade, it's also important to select an underlying with an appropriate amount of risk for a given account size. The two broad classes of instruments in the example asset universe, stocks and ETFs generally have different volatility profiles, and there are pros and cons to trading each, summarized in [Table 5.2](#).

Table 5.2 General pros and cons for stock and ETF underlyings.

Stocks		ETFs	
Pros	Cons	Pros	Cons
• Tend to have options with higher credits and higher profit potentials • Frequent high implied volatility (IV) conditions	• Single-company risk factors • Earnings and dividend risk • Tend to have options with higher buying power reductions (BPRs)	• Inherently diversified across sectors or markets • Tend to have options with lower BPRs and are still highly liquid	• Limited selection compared to stocks • High IV conditions are not common

When choosing an underlying, the capital requirement of the trade is a limiting factor. A single position should generally occupy no more than 5% to 7% of portfolio capital, meaning that stock underlyings may not be suitable for small accounts because they are more expensive to trade. However, since selling premium when IV is elevated has several benefits, stock underlyings may be preferable underlyings in certain circumstances. As stocks are subject to company- and sector-specific risks, they tend to have higher IVs compared to ETFs and tend to present elevated IV

opportunities more often. Note that if trading stock options, investors should also be aware of the contextual information (e.g., earnings reports dates, company announcements) that may be driving these periods of IV inflation because it may impact the strategy choice.¹ This practice is less important when trading options with ETF underlyings.

The additional risk factors (coupled with the fact that liquid stocks are often more expensive than ETFs) result in stock options generally having much larger profit and loss (P/L) swings throughout the contract duration, more ending P/L variability, and more tail risk. If the capital requirements of the trade are not excessive and the IV of the underlying is favorable, then these will be the next factors to consider. Overall, stock options are usually riskier but also carry a higher profit potential than ETF options. Consider the statistics outlined in [Table 5.3](#).

Table 5.3 Options P/L and probability of profit (POP) statistics 45 days to expiration (DTE) 16 Δ strangles with six different underlyings, held to expiration, from 2009–2020. Assets include SPY (S&P 500 ETF), GLD (gold commodity ETF), SLV (silver commodity ETF), AAPL (Apple stock), GOOGL (Google stock), and AMZN (Amazon stock).

16 Δ Strangle Statistics, Held to Expiration (2009–2020)				
	Underlying	Average Profit	Average Loss	POP
ETFs	SPY	\$160	–\$297	82%
	GLD	\$125	–\$424	83%
	SLV	\$33	–\$103	81%
Stocks	AAPL	\$431	–\$1,425	76%
	GOOGL	\$1,108	–\$2,886	80%
	AMZN	\$1,041	–\$2,215	78%

The tolerance for P/L swings, ending P/L variability, and tail exposure depends mostly on account size and personal risk preferences. If a trade approximately satisfies those preferences and the constraints previously stated, then the choice of underlying is somewhat irrelevant because of a concept called product indifference. Because IV is derived from option price, if two assets have the same IV, their options will have roughly the

same price (as a percentage of underlying price). Consequently, one underlying will not give more edge with respect to options pricing inefficiencies compared to another, provided they have similarly liquid options markets. To visualize this, consider the example in [Table 5.4](#).

Table 5.4 Two sample options underlyings with the same IV but differing stock and put prices.

Option Parameters	Scenario A	Scenario B
Stock Price	\$100	\$200
IV	33%	33%
45 DTE 16 Δ Put Price	\$1	\$2

In Scenario A, the put is \$1 (1% of the underlying price). Due to the efficient nature of options pricing, the short put in Scenario B will also cost 1% of the underlying price, as both assets have the same IV. Product indifference suggests that no one (liquid) underlying is inherently superior to another, merely that there are proportional trade-offs among different assets. The high-risk, high-reward nature of stocks is not inherently better or worse than the relatively stable nature of ETFs, but some assets may be more suitable for an individual trader than others. We can, thus, conclude that the choice of an underlying essentially depends on five main factors (in order of significance):

1. The liquidity of the options market
2. The BPR of the trade relative to account size²
3. The IV of the underlying³
4. The preferred magnitude of P/L swings, ending P/L variability, and tail exposure per trade
5. The preferred company, sector, or market exposure

Choose a Contract Duration

There are many ways to choose a contract duration, but this book approaches this process from a qualitative perspective. The three primary

goals when choosing a contract duration are summarized as follows:

1. Using portfolio buying power effectively.
2. Maintaining consistency and reaching a large number of occurrences.
3. Selecting a suitable time frame given contextual information.

It is essential to determine what contract duration is the most effective use of portfolio buying power without exceeding risk tolerances. Premium prices tend to be more sensitive to changes in underlying price (higher gamma) for contracts that are near expiration (5 DTE) compared to contracts that are far from expiration (120 DTE). Consequently, short-term contracts tend to have significant P/L swings for a larger portion of their duration compared to longer-term contracts, which initially have more moderate P/L swings and gradually become more volatile. Most contracts also tend to exhibit an increase in P/L instability as they near expiration, which is also a consequence of higher gamma. The gamma of a contract tends to increase throughout a contract's duration, usually the result of the underlying price drifting closer to one of the strangle strikes over time.

[Figure 5.1](#) illustrates these concepts by comparing the standard deviation of daily P/Ls for different durations of the same type of contract.

All of these strangles exhibit a decrease in P/L swings right before expiration. This is because options rapidly lose their extrinsic value near expiration, presuming they are not in-the-money (ITM), which is usually the case because 16 Δ options often expire worthless. Near expiration, this exponential decline in premium from theta decay outweighs the magnitude of the P/L swings.

The P/L swings at the beginning of the contract vary greatly based on the contract duration. On day seven, the daily P/L for the 15 DTE contract has a high variance, and the 30 DTE, 45 DTE, and 60 DTE contracts have much lower P/L swings around the seven-day mark. This is because the 16 Δ strikes in the 15 DTE contract are much closer to the at-the-money (ATM) than the 16 Δ strikes in the 30+ DTE contracts. This is shown numerically in [Table 5.5](#).⁴

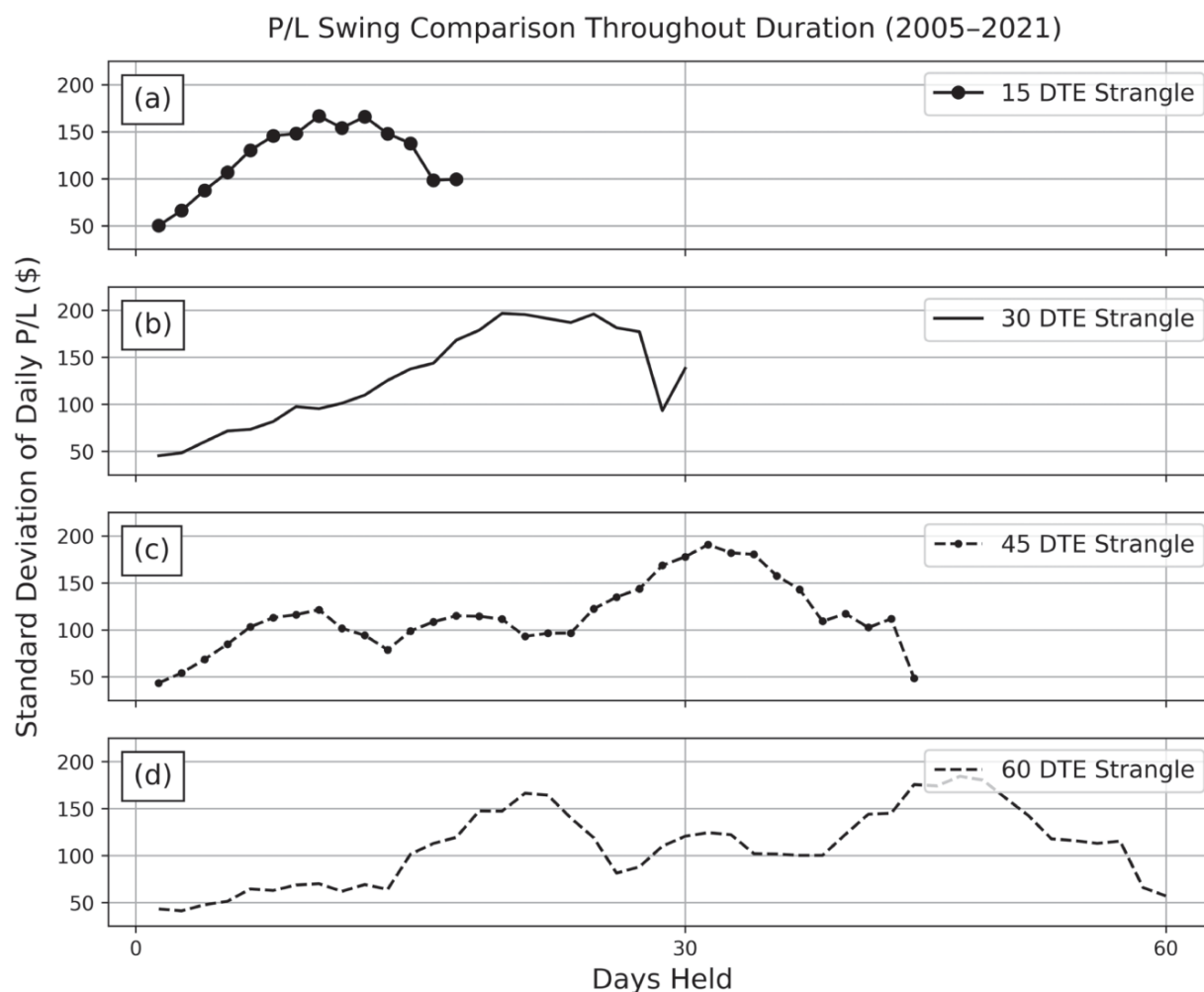


Figure 5.1 Standard deviation of daily P/Ls (in dollars) for 16 Δ SPY strangles with various durations from 2005–2021. Included are durations of (a) 15 DTE, (b) 30 DTE, (c) 45 DTE and (d) 60 DTE.

[Table 5.5](#) illustrates how the 16 Δ strikes are closer to the stock price for the 15 DTE contract compared to longer duration strangles. Therefore, small changes in the price of the underlying will have a larger impact on the option's delta compared to contracts with longer durations and further out strike prices. The 30+ DTE contracts tend to experience larger P/L swings once they near expiration because the underlying price often drifts toward one of the strikes over time.

Longer contract durations, because their P/L swings are manageable for a longer period of time, give traders more time to assess the trade and adjust to changes in market conditions. However, trade durations that are too long are not necessarily effective uses of buying power because they do not

allow for as many occurrences and take a longer time to generate profits. To summarize, longer-term contracts, which don't typically experience large changes in P/L until the latter half of their duration, tie up buying power for a long time without generating significant profit most of that time. By comparison, shorter-term contracts exhibit volatile P/L swings for the majority of their duration and leave little time to react to new conditions. A middle ground contract duration, one between 30 and 60 days on a monthly expiration cycle,⁵ is considered a suitable use of buying power. Middle ground durations offer a period of manageable P/L swings while providing a fair amount of daily premium decay and a reasonable timescale for profit. This buffer time allows traders to evaluate the viability of a trade before P/L swings become more volatile. It also allows traders to incorporate different trade management strategies, which will be covered in the next chapter.

Table 5.5 Data for 16 Δ SPY strangles with different durations from April 20, 2021. The first row is the distance between the strike for a 16 Δ put and the price of the underlying for different contract durations (i.e., if the price of the underlying is \$100 and the strike for a 16 Δ put is \$95, then the put distance is $(\$100 - \$95)/\$100 = 5\%$). The second row is the distance between the strike for a 16 Δ call and the price of the underlying for different contract durations.

16 Δ SPY Option Distance from ATM			
Option Type	15 DTE	30 DTE	45 DTE
Put Distance	3.9%	6.5%	8.0%
Call Distance	2.4%	3.9%	4.9%

Another important factor to consider when choosing a contract duration is consistency and the number of occurrences. Consistently choosing similar contract time frames increases the number of occurrences and simplifies portfolio management because expiration and management times will roughly align for the majority of short premium trades in a portfolio. As discussed in [Chapter 3](#), a large number of occurrences is required to reduce the variance of portfolio averages and maximize the likelihood of realizing long-term expected values. For profit and risk expectations to be dependable, it is essential to choose contract durations (and management strategies) that allow for a reasonable number of occurrences and to do so

consistently. Therefore, it's good practice to choose a contract time frame that is convenient to maintain and short enough to allow for several trades to be placed throughout the trading year, presuming the duration maintains a manageable amount of tail risk exposure.

The final major factor when choosing a contract duration is contextual information, particularly when trading stock options. Contextual information, such as an approaching election, earnings report date, or forecasted natural disaster cannot necessarily be used to consistently forecast price direction, but it may indicate a predictable change in price volatility. There is, therefore, utility in taking contextual information into account when choosing a contract time frame. This will be discussed in more detail in [Chapter 9](#).

Choose Defined or Undefined Risk

Long options strategies are defined risk trades, as the maximum loss is capped by the price of the contract. Short options positions may have defined or undefined risk profiles. Defined risk strategies have a fixed maximum loss, but capping downside risk has drawbacks. Undefined risk strategies have unlimited downside risk, meaning the maximum loss on a trade-by-trade basis is potentially unlimited. The pros and cons of defined and undefined risk strategies are outlined in [Table 5.6](#).

Table 5.6 Comparison of defined and undefined risk strategies.

Undefined Risk		Defined Risk	
Pros	Cons	Pros	Cons
• Higher POPs • Higher profit potentials	• Unlimited downside risk • Higher BPRs (more expensive to trade)	• Limited downside risk • Lower BPRs (less expensive to trade)	• Lower POPs • Lower profit potentials • Can run into liquidity issues^a

^aDefined risk trades, because they consist of short premium and long premium contracts, require more contracts to be filled than equivalent undefined risk trades. There is, therefore, a higher risk that a defined risk order will be unable to close at a good price compared to an equivalent undefined risk position.

Defined risk strategies have a known maximum loss (i.e., the BPR of the trade) and will typically have a lower BPR than an undefined risk strategy with similar parameters (underlying, contract duration, strikes). Although defined risk positions expose less capital than equivalent undefined risk positions, this does not imply they carry less risk.

Recall from the discussion of option risk in [Chapter 3](#) that there are several ways to quantify the risk of an options strategy. Though defined risk strategies avoid carrying *outlier risk*, they are more likely to lose most or all their BPR when losses do occur. It's, therefore, *essential* to recognize that BPR is mathematically and functionally different for defined and undefined risk trades, and it *cannot* be used as a comparative risk metric between them. This will be discussed later in the chapter.

Due to the differences in POP and profit potential between risk profiles, the maximum amount of portfolio capital allocated should differ depending on whether the strategy is defined or undefined risk. For undefined risk strategies, traders are compensated for the significant tail risk with high

profit potentials and high POPs. It is generally recommended that undefined risk strategies constitute the majority of portfolio capital allocated to short premium strategies. More specifically, *at least* 75% of allocated capital should be in undefined risk strategies (with a maximum of 7% allocated per trade) and at most 25% of allocated capital should be in defined risk strategies (with a maximum of 5% allocated per trade). For a numerical example, consider the allocation scenarios for a \$100,000 portfolio described in [Table 5.7](#).

Table 5.7 Portfolio allocation for defined and undefined risk strategies with a \$100,000 portfolio at different VIX levels.

VIX Level	Maximum Portfolio Allocation	Minimum Undefined Risk Allocation	Maximum Defined Risk Allocation
20	\$30,000	\$22,500 (\$7,000 max BPR per trade)	\$7,500 (\$5,000 max BPR per trade)
40	\$50,000	\$37,500 (\$7,000 max BPR per trade)	\$12,500 (\$5,000 max BPR per trade)

These differences will be elaborated on in the next section, but to summarize, the following five factors are generally the most important to consider when comparing defined and undefined risk trades:

1. The amount of BPR required for a trade relative to the net liquidity of the portfolio.
2. The likelihood of profiting from a position.
3. The preferred amount of downside risk.
4. The preferred ending P/L variability and preferred magnitude of P/L swings throughout the contract duration.
5. The profit targets.

Defined risk trades typically require less capital, have more moderate P/L swings throughout the trade, and have less ending P/L standard deviation compared to undefined risk trades. Consequently, defined risk trades may be preferable for small accounts and relatively new traders. Undefined risk

trades are statistically favorable and have, therefore, been the focus of this book. However, the following section discusses how to construct defined risk trades that behave like undefined risk trades while offering protection against extreme losses. For these types of strategies, and only these types of strategies, defined risk trades can be substituted for undefined risk trades in the portfolio allocation guidelines.

Choose a Directional Assumption

After choosing a contract underlying, duration, and risk profile, the next steps are determining the directional assumption for the price of the underlying asset and selecting a strategy consistent with that belief and the preferred risk profile. The directional assumption may be bullish, bearish or neutral, and the optimal choice is subjective and depends on one's interpretation of the efficient market hypothesis (EMH). Recall that the EMH assumes that current prices reflect some degree of available information and comes in three main forms:

1. Weak EMH: Current prices reflect all past price information.
2. Semi-strong EMH: Current prices reflect all publicly available information.
3. Strong EMH: Current prices reflect all possible information.

Each form of the EMH implies some degree of limitation with respect to price predictability:

- Weak EMH: Past price information cannot be used to consistently predict future price information, which invalidates technical analysis.
- Semi-strong EMH: Any publicly available information cannot be used to consistently predict future price information, which invalidates fundamental analysis.
- Strong EMH: No information can be used to consistently predict future price information, which invalidates insider trading.

No form of the EMH is universally accepted or rejected, and the ideal form to trade under (if any) depends on personal preference. This book takes a

semi-strong approach to market predictability, assuming equity and option prices effectively reflect available information and that few directional assumptions are valid (e.g., the market trends bullish in the long term). As volatility reverts back to a long-term value following significant deviations, it is more valid to make directional assumptions on IV once it's inflated rather than directional assumptions around equity prices. This book, therefore, typically focuses on directionally *neutral* strategies, such as the short strangle, because these types of positions profit from changes in volatility and time and are relatively insensitive to price changes. However, that is a personal choice. Multiple strategies are outlined in [Table 5.8](#).

For reasons discussed in earlier chapters, all these strategies perform best in high IV. However, the POPs of these trades remain relatively high in all volatility environments, justifying that some percentage of capital should be allocated in all IV conditions.

To elaborate on the differences between defined and undefined risk, compare statistics for the two neutral strategies: the iron condor and the strangle. An iron condor consists of a short out-of-the-money (OTM) vertical call spread and a short OTM vertical put spread (introduced in [Table 5.8](#)). This trade is effectively a short strangle paired with a long strangle having strikes that are further OTM (typically called wings). As with strangles, iron condors are profitable when the underlying price stays within the range defined by the short strikes or when there is a significant IV contraction or time decay. For example, a 16 Δ strangle can be turned into a 16 Δ iron condor with 10 Δ wings⁶ with the addition of a long call and a long put with the same duration, further from OTM (the long contracts are 10 Δ in this case). An example of an iron condor is shown in [Table 5.9](#) and [Figure 5.2](#).

Table 5.8 Examples of popular short options strategies with the same delta of approximately 20.^a

Strategy	Composition	Defined or Undefined Risk	Directional Assumption	POP ^b
Naked Option	Short OTM put	Undefined	Bullish	80%
	Short OTM call	Undefined	Bearish	80%
Vertical Spread	Short OTM put, long further OTM put	Defined	Bullish	77%
	Short OTM call, long further OTM put	Defined	Bearish	77%
Strangle	Short OTM put, short OTM call	Undefined	Neutral	70%
Iron Condor	Short OTM vertical call spread, short OTM vertical put spread	Defined	Neutral	60%

^aThe directional assumption will be flipped for the long side of a non-neutral position. For a long neutral position, the assumption is that the underlying price will move outside of the price range defined by the contract strikes. The POP of the long side is given by $1 - (\text{short POP})$.

^bThese POPs are approximate. The POP of a defined risk strategy depends heavily on the choice of long delta(s). Contracts with wider long deltas will generally have higher POPs. This will be explored more later in the chapter.

The long wings of the iron condor cap the maximum loss as either the difference between the strike prices of the vertical put spread or vertical call spread (whichever is greater) times the number of shares in the contract (typically 100) minus the net credit. The maximum loss of the short iron condor is equivalently the BPR required to open the trade. It can be summarized by the following formula:

Short Iron Condor BPR (5.1)

$$\begin{aligned}
 &= 100 \times \max(\text{long call strike} - \text{short call strike}, \\
 &\quad \text{short put strike} - \text{long put strike}) \\
 &\quad - 100 \times (\text{short call price} + \text{short put price} \\
 &\quad - \text{long call price} - \text{long put price})
 \end{aligned}$$

Continuing with the same example as shown in [Table 5.9](#), we apply this formula to calculate some statistics for these two trades in [Table 5.10](#).

Table 5.9 Example of a 16 Δ SPY strangle and a 16 Δ SPY iron condor with 10 Δ wings when the price of SPY is \$315 and its IV is 12%. All contracts must have the same duration.

Contract Strikes	16 Δ Strangle	16 Δ Iron Condor with 10 Δ Wings
Long Call Strike	---	\$332
Short Call Strike	\$328	\$328
Short Put Strike	\$302	\$302
Long Put Strike	---	\$298

The short strikes were approximated with the expected range formula and the long strikes for the iron condor wings were approximated with the Black-Scholes formula. Underlyings are often subject to strike skew, not to be confused with distribution skew, which neither of these methods really consider. This means that the strikes (both long and short) are typically not equidistant (as a dollar amount) from the price of the underlying although they were approximated in this example as such. This concept will be explored more later in the chapter.

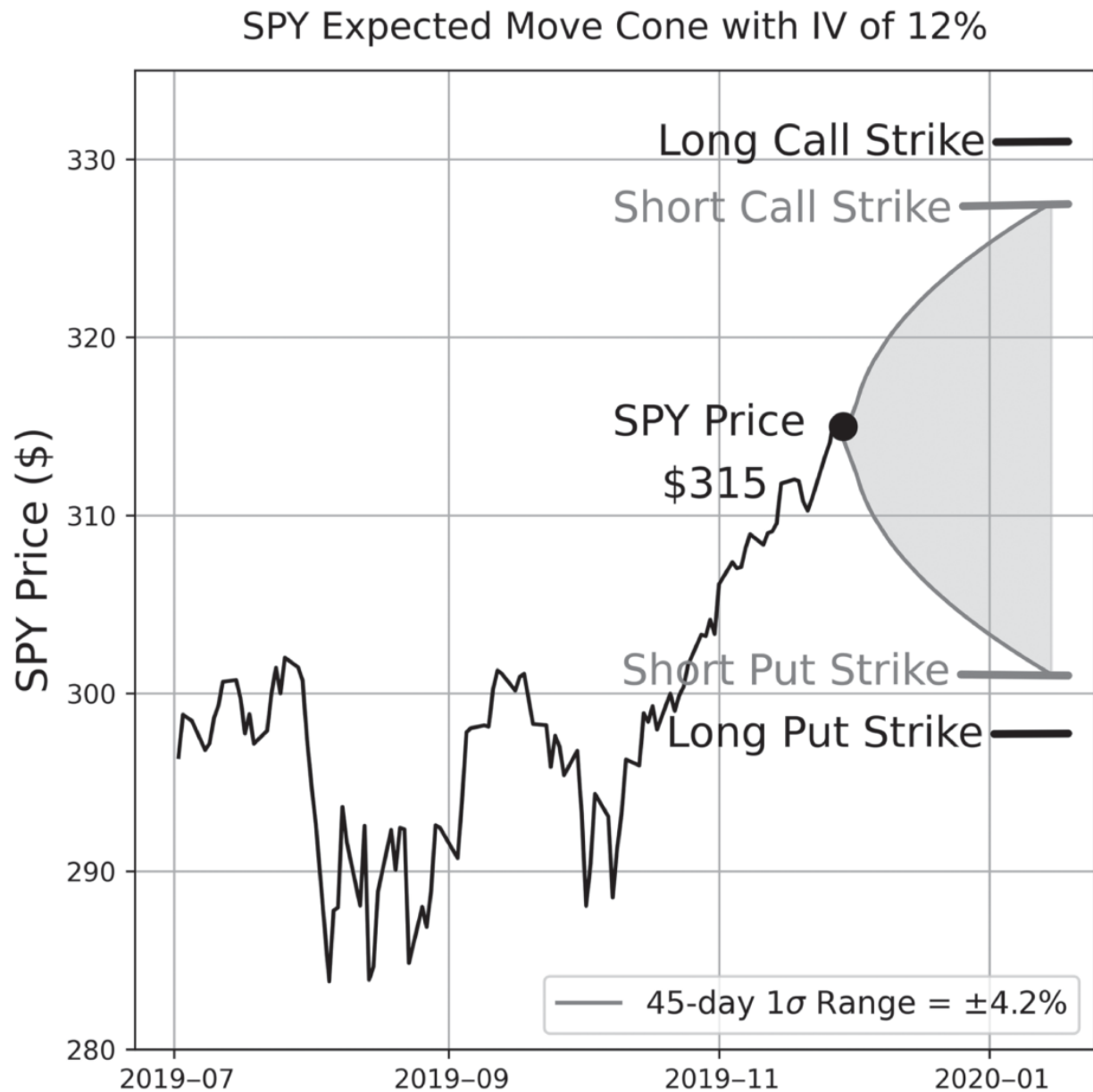


Figure 5.2 Graphical representation of the iron condor described in [Table 5.9](#). The 10 Δ wings correspond to long strikes that are \$17 from ATM, which is further OTM than the 16 Δ short strikes that are \$13 from ATM.

Table 5.10 Initial credits for the 16 Δ SPY strangle and the 16 Δ SPY iron condor with 10 Δ wings outlined in [Table 5.9](#). Because the difference between the vertical call spread strikes (\$332–\$328) and the vertical put spread strikes (\$302–\$298) is the same (\$4), this value is used when calculating the maximum loss.

Contract Credits	16 Δ Strangle	16 Δ Iron Condor with 10 Δ Wings
Long Call Debit	---	–\$69
Short Call Credit	\$122	\$122
Short Put Credit	\$108	\$108
Long Put Debit	---	–\$57
Net Credit	\$230	\$104
Max Loss	∞	$(100 \text{ shares} \times \\$4) - \\$104 = \\296
BPR	\$5,000	\$296

The choice of wing width depends on personal profit targets and the threshold for extreme loss. Large losses generally occur once the long put or call strikes are breached by the price of the underlying, so wings that are further from ATM are exposed to larger outlier moves but are more likely to be profitable. Wings that are closer to ATM are more expensive but also reduce the maximum loss of a trade. To summarize, wings that are further out yield iron condors with a larger profit potential and a higher probability of profit but also larger possible losses. For some numerical examples, refer to the statistics in [Table 5.11](#).

Table 5.11 Statistical comparison of 45 DTE 16 Δ SPY iron condors with different wing widths, held to expiration from 2005–2021. Wings that have a smaller delta are further from ATM compared to wings with a larger delta. Included are 45 DTE 16 Δ SPY strangle statistics held to expiration from 2005–2021 for comparison.

16 Δ Iron Condor Statistics (2005–2021)				16 Δ Strangle Statistics (2005–2021)
Statistics	5 Δ Wings	10 Δ Wings	13 Δ Wings	
POP	79%	75%	73%	81%
Average P/L	\$35	\$15	\$6	\$44
Standard Deviation of P/L	\$251	\$132	\$74	\$614
Conditional Value at Risk (CVaR) (5%)	–\$771	–\$399	–\$220	–\$1,535

If the account size allows for it, it is preferable to trade iron condors with *wide wings*, which have more tail risk than narrow iron condors but are historically more profitable. While iron condors with narrow wings have POPs near 70%, wide iron condors may have POPs of nearly 80%, as shown in [Table 5.11](#). Wider iron condors, although they have higher BPR requirements, are also less likely to reach max loss than tighter iron condors when losses do occur.

Defined risk strategies tend to have lower POPs and profit potentials compared to undefined risk strategies as shown by the strangle statistics included for reference. The iron condor has roughly a third of the profit potential as the strangle on average (in the case of 10 Δ wings), but it also has roughly a third of the P/L standard deviation and significantly less tail exposure. Also, as in the wide iron condor example, defined risk trades can be constructed to have similar POPs as an undefined risk strategy while still offering protection from outlier losses. Defining risk in low IV, particularly with strategies that have high POPs, is one way to manage the outlier loss exposure while capitalizing on the benefits of short premium. Defined risk strategies also come with the added benefit of being significantly cheaper to trade, which is another reason they may be a more effective use of portfolio

buying power when IV is low. For a numeric reference, consider the BPR statistics in [Table 5.12](#).

Table 5.12 Average BPR comparison of 45 DTE 16 Δ SPY strangles and 45 DTE 16 Δ SPY iron condors with 10 Δ wings when held to expiration using data from 2005–2021.

SPY Strangle and Iron Condor BPRs (2005–2021)		
VIX Range	Strangle BPR	Iron Condor BPR ^a
0–15	\$3,270	\$363
15–25	\$2,641	\$426
25–35	\$2,261	\$585
35–45	\$1,648	\$553
45+	\$1,445	\$615

^aIron condors with static dollar wings (e.g., \$10 wings, \$20 wings) have BPRs that decrease with IV as seen with strangles. Iron condors with dynamic wings that change with variables, such as IV (e.g., 10 Δ , 5 Δ) have BPRs that increase with IV. Recall that the iron condor BPR is the widest short spread width minus the initial credit. Therefore, as IV increases, the widest width increases faster than the initial credit, so the BPR increases with IV.

Defined risk strategies *with high POPs* can account for a greater percentage of portfolio allocation than defined risk strategies with lower POPs.

Previously, we stated that at least 75% of allocated capital should be in undefined risk strategies (with a maximum of 7% allocated per trade) and at most 25% of allocated capital should be in defined risk strategies (with a maximum of 5% allocated per trade). However, a defined risk strategy with a POP comparable to an undefined risk strategy can share undefined risk portfolio buying power, which protects capital from extreme losses while still allowing for consistent profits. Once IV expands, traders can then transition to strangles to capitalize on the higher credits and reduced outlier risk.

It's crucial to reiterate that BPR *cannot* be used to compare risk between strategies with different risk profiles. For instance, refer back to the example in [Table 5.10](#). The strangle requires roughly 17 times more buying power than the iron condor, but this is not to say that the risk of the strangle is equivalent to the risk of 17 iron condors. The strangle is more likely to be

profitable and much less likely to lose the entire BPR because that would require a much larger move in the underlying (20%) compared to the iron condor (5%). Very wide iron condors have similar risk profile to strangles, but it is generally good practice to avoid directly comparing defined and undefined risk strategies using buying power.

Choosing a Delta

Recall that delta is a measure of *directional exposure*. According to the mathematical definition derived from the Black-Scholes model, it represents the expected change in the option price given a \$1 increase in the price of the underlying (assuming all other variables stay constant).⁷ For example, if the price of an underlying increases by \$1, the price for a long call option with a delta of 0.50 (denoted as 50 Δ) will increase by approximately \$0.50 per share, and the price for a long put option with a delta of -0.50 (denoted as -50Δ , or just 50Δ when the sign is clear from context) will decrease by approximately \$0.50 per share.⁸ The delta of a contract additionally represents the *perceived* risk of that option in terms of shares of equity. More specifically, delta corresponds to the number of shares required to hedge the directional exposure of that option according to market sentiment.

This book frequently references the 16 Δ SPY strangle, which is a delta neutral trade consisting of a short 16 Δ put directionally hedged with a short 16 Δ call. Delta neutral positions profit from factors such as decreases in IV and time decay rather than directional changes in the underlying. When originally presented in [Chapter 3](#), the short strike prices were related to the expected range, and therefore, strike prices were shown to be equidistant from the price of the underlying as in [Figure 5.3](#).

The strikes in this example were derived from the expected move range approximation shown in [Chapter 2](#). However, in practice the strikes for a 16 Δ SPY put/call are calculated from real-time supply and demand and are often subject to *strike skew*. Revisit the example from [Table 5.5](#) to see an example of this.

[Table 5.5](#) shows that the put strikes are much further from the price of the underlying compared to the call strikes even though the call and put

contracts are both 16 Δ . According to market demand, put contracts further OTM have equivalent risk as call contracts less OTM. This skew results from market fear to the *downside*, meaning the market fears larger extreme moves to the downside more than extreme moves to the upside.⁹ As delta is based on the market's perception of risk, strikes for a given delta are skewed according to that perception. But not all instruments will have the same degree of skew. Stocks like AAPL and GOOGL have fairly equidistant strikes, but market indexes and commodities (e.g., gold and oil) tend to have downside skew, otherwise known as put skew. Assets like GME (GameStop) and AMC (entertainment company) developed upside skew, otherwise known as call skew, during 2020.

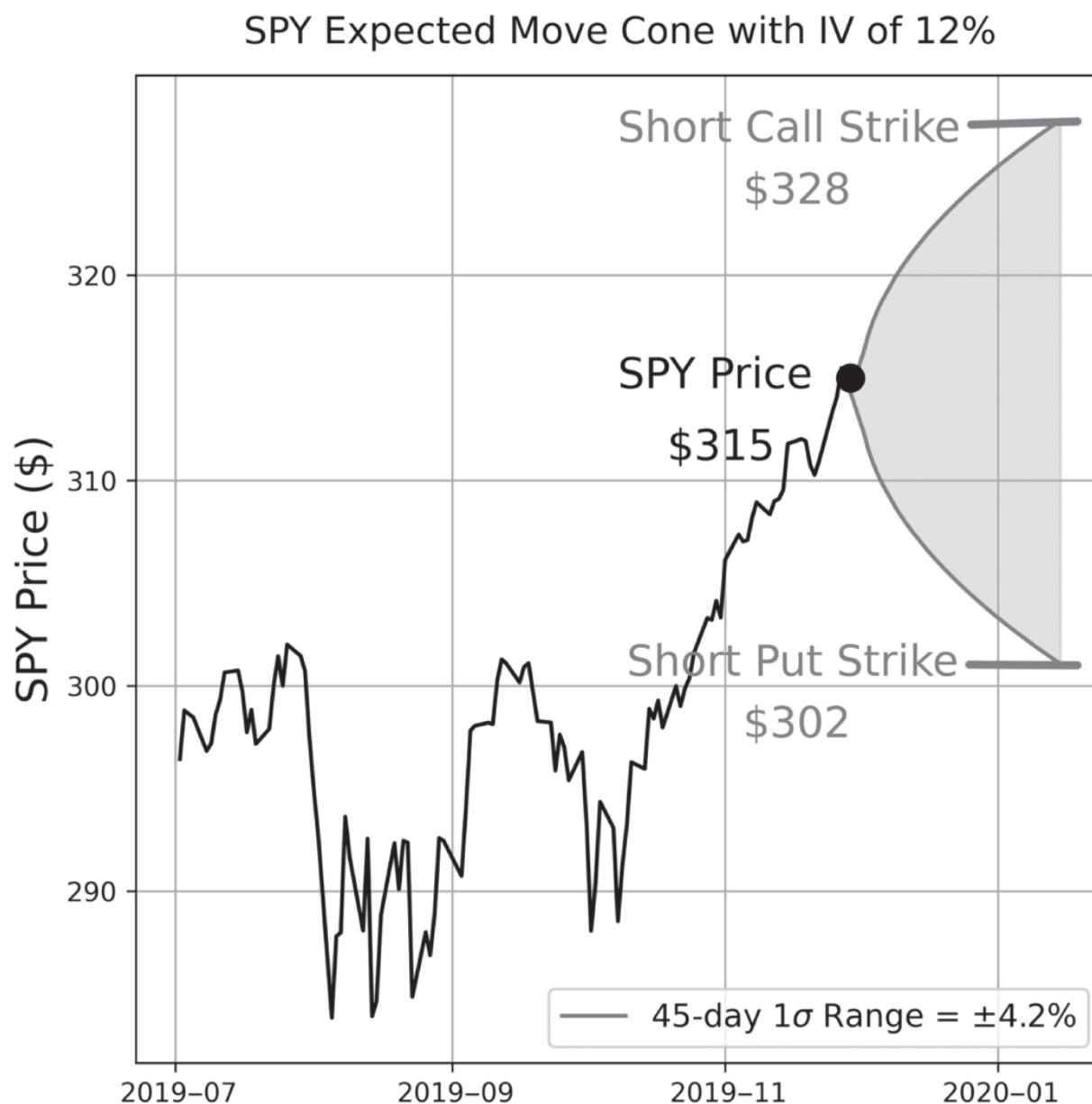


Figure 5.3 The price of SPY in the last 5 months of 2019. Included is the 45-day expected move cone calculated from the IV of SPY in December 2019, where the strike for the 16 Δ call is \$328 and the strike for the 16 Δ put is \$302.

Because delta is a measure of perceived risk in terms of share equivalency, the chosen delta is going to significantly impact the risk-reward profile of a trade. Positions with larger deltas (closer to -100Δ or $+100 \Delta$) are more sensitive to changes in the price of the underlying compared to positions with smaller deltas (closer to 0). To observe how this impacts per-trade

performance, consider the statistics for 45 DTE SPY strangles with different deltas outlined in [Tables 5.13–5.15](#).

Table 5.13 Statistical comparison of 45 DTE SPY strangles of different deltas, held to expiration from 2005–2021.

SPY Strangle Statistics (2005–2021)			
Statistics	16 Δ	20 Δ	30 Δ
POP	81%	76%	68%
Average P/L	\$44	\$49	\$54
Standard Deviation of P/L	\$614	\$659	\$747
CVaR (5%)	–\$1,535	–\$1,673	–1,931

Table 5.14 Average BPRs of 45 DTE SPY strangles with different deltas, sorted by IV from 2005–2021.

SPY Strangle BPRs (2005–2021)			
VIX Range	16 Δ	20 Δ	30 Δ
0–15	\$3,270	\$3,366	\$3,573
15–25	\$2,641	\$2,756	\$3,014
25–35	\$2,261	\$2,415	\$2,794
35–45	\$1,648	\$1,715	\$2,058
45+	\$1,445	\$1,421	\$1,520

Table 5.15 Probability of incurring a loss exceeding the BPR for 45 DTE SPY strangles of different deltas, held to expiration from 2005–2021.

SPY Strangle Statistics (2005–2021)	
Strangle Delta	Probability of Loss Greater Than BPR
16 Δ	0.90%
20 Δ	0.93%
30 Δ	1.0%

Positions with higher deltas have larger P/L swings throughout the contract duration, more ending P/L variability, higher BPRs and lower POPs compared to positions with lower deltas. However, higher delta positions also carry higher credits and larger profit potentials overall. Positions with lower deltas achieve smaller profits more often and are lower risk than higher delta trades. Positions with lower deltas also tend to have higher thetas as a percentage of the option value, meaning they may reach profit targets more quickly than positions with higher deltas (not shown in these tables).

The optimal choice of delta depends on the personal profit goals and, most importantly, personal risk tolerances. ITM options (options with a delta magnitude larger than 50) generally carry more directional risk and an insufficient amount of theta (expected daily profits due to time decay) than what is suitable for a short premium trade. OTM options are typically better candidates. When trading short premium, contract deltas between 10 Δ and 40 Δ are typically large enough to achieve reasonable growth but small enough to have manageable P/L swings, moderate standard deviation of ending P/L, and moderate outlier risk. More risk-tolerant traders generally trade options over 25 Δ and more risk-averse traders will trade under 25 Δ . When IV increases and options become cheaper to trade, more risk-tolerant traders may also scale delta *up* to capitalize on the larger credits across the entire options chain. It is also good practice to re-center the deltas of existing positions when IV increases because increases in IV cause the strike price for a given delta to move *further away* from the spot price. To see an example of this, consider [Table 5.16](#).

Table 5.16 Comparison of strike prices for two 30 DTE 16 Δ call options with the same underlying price but different IVs.

Example Parameters for a 30 DTE 16 Δ Call Option		
IV	Underlying Price	Strike Price
10%	\$100	\$103
50%	\$100	\$117

The strike price for a 16 Δ call is \$17 away from the price of the underlying when the IV is 50%, compared to \$3 away when the IV is 10%.

This is because an increase in IV indicates an increase in the expected range for the underlying price. When this expected range becomes larger, contracts with strikes further away from the current price of the underlying are in higher demand than in lower IV conditions. This demand increases the premiums of those contracts and consequently the perceived risk. When IV increases, it is good practice to close existing positions and reopen them with adjusted strikes that better reflect the new volatility conditions.

Takeaways

1. Constructing a trade has six major steps, and the ideal choices are based on account size and the personal profit goals, risk tolerances, and market assumptions. The primary factors to consider are the asset universe, the underlying, the contract duration, the risk profile of the strategy, the directional assumption, and the delta.
2. Traders should choose assets with highly liquid options markets, consisting of contracts that can be easily converted into cash without a significant impact on market price. Liquid options markets have a high volume across strikes, tight bid-ask spreads, and available contracts with several strike prices and expiration dates.
3. In an equity-focused asset universe, traders have two main choices of equity underlyings: stocks and ETFs. Options with stock underlyings tend to have higher credits, higher profit potentials, and more frequent high IV conditions, but they also have single-company risks and cost more to trade than options with ETF underlyings. ETFs are inherently diversified and are cheaper than stocks while being very liquid, but fewer choices are available and high IV conditions are less common.
4. A suitable contract duration should use buying power effectively, allow for consistency and a reasonable number of occurrences, and reflect the timescale of contextual events, such as upcoming earnings reports and forecasted natural disasters. Contract durations ranging from 30 to 60 days are generally a suitable use of portfolio buying power, offering manageable P/L volatility and a reasonable timescale for profit.

5. Short premium strategies may have defined or undefined risk. Undefined risk trades have higher POPs and higher profit potentials but also unlimited downside risk and higher BPRs, making them more expensive to trade. Defined risk strategies have limited downside risk and lower BPRs but also lower POPs and lower profit potentials with possible liquidity issues. High-POP defined risk strategies, such as wide iron condors, can occupy the capital reserved for undefined risk trades, and this is a particularly good strategy when IV is low. Trading high-POP defined risk trades in low IV and transitioning to undefined risk in high IV is an effective way to protect capital from outlier moves while profiting consistently.
6. Traders must choose one of three directional assumptions for the underlying price: bullish, bearish, and neutral. The optimal choice is subjective and depends on individual interpretation of the EMH, which assumes current prices reflect some degree of available information.
7. The delta of a contract represents the perceived risk of the option in terms of shares of equity, making the choice of delta based on personal risk tolerances and profit goals. A higher delta OTM contract is closer to ATM and more sensitive to changes in underlying price, meaning that these positions are generally riskier but have higher profit potentials. Lower delta OTM contracts are further from ATM and have more moderate P/L swings throughout the contract with lower ending P/L standard deviation generally. When trading short premium, ITM contracts are generally not suitable due to their high directional risks and low thetas. Contracts between 10 Δ and 40 Δ are generally large enough to achieve a reasonable amount of growth but small enough to have manageable P/L swings and moderate ending P/L variability.

Notes

- 1 IV inflation specifically due to earnings is the basis for a type of strategy called an earnings play. Earnings plays will be discussed in [Chapter 9](#) and for now will not be part of stock options discussions.
- 2 This will be explored more later in this chapter and in [Chapter 7](#), when covering the portfolio allocation guidelines in more detail.

- 3 In practice, IV is often interpreted according to the IV percentile or IV rank of the underlying. This is a more common trading metric because traders are rarely deeply familiar with the IV dynamics of different assets, and it is essential to include a range of assets in a balanced portfolio.
- 4 The put distance and call distance are not symmetric. This is due to strike skew, which will be discussed later in this chapter and in the appendix.
- 5 Common options expiration dates are divided into weekly, monthly, and quarterly cycles. Contracts with *monthly* expirations cycles are preferable because they are consistently liquid across liquid underlyings. For highly liquid assets, any expiration cycle is acceptable.
- 6 Recall that smaller deltas are further from ATM than larger deltas.
- 7 For contracts with deltas between approximately 10 and 40, delta can also be used as a *very* rough proxy for the probability that an option will expire ITM. For instance, a 25 Δ put has about a 25% chance of expiring ITM, meaning that there is a 75% POP for the short put. A 16 Δ strangle is composed of a 16 Δ put and a 16 Δ call, so there is approximately a 32% chance that it will expire ITM (consistent with the 68% POP for the short strangle).
- 8 Delta is between 0 and 1 for long calls and between -1 and 0 for long puts. For short calls and short puts, the numbers are flipped.
- 9 This is mainly the result of the history of extreme market crashes, such as the 1987 Black Monday crash, the 2008 housing crisis, and the 2020 sell-off. Prior to 1987, the put and call strikes of the same delta were much closer to equidistant.